

MetaboTyping RA Agent Workbench

Metabolomics Workbench census and RA Agent capabilities

Condensed evidence report | MW corpus snapshot: 24 August 2026, 04:29 UTC

MW counts in this report come from one frozen, provenance-tracked retrieval of the public bulk indexes. The report also shows how the MetaboTyping RA Agent Workbench can use those records. Study summaries, analyses, organism incidences, reported feature counts, and RefMet nomenclature entries remain separate because they have different denominators.

Measure	Result	Interpretation
MW study summaries	4,499	Unique ST records after one additional row sharing a stud...
MW analyses	7,367	Unique analysis IDs after three additional rows sharing a...
Organism coverage	4,483 studies; 4,770 incidences; 481 labels	4,395 single-species and 88 multi-species studies
Assay evidence	2,385 untargeted; 14 explicit targeted; 4,968 unclassifi...	Analysis-level categories; targeted is a strict lower bou...
Reported feature volume	1,231,376 instances across 5,420 numeric records	Median 99 per reported analysis; not unique metabolites
RefMet reference snapshot	205,948 entries	23 superclasses, 333 main classes, 960 subclasses
Workbench inventory	22 canonical roles; 24 logical skills	Two deprecated role aliases are not extra capabilities
Source-routing inventory	28 sources; 27 lanes	Registration does not imply that every source has a live ...

1. Metabolomics Workbench corpus

Study and species coverage

The official MW bulk endpoints returned 4,500 summary rows and 7,370 analysis rows. We kept the first record for each identifier. One additional summary row shared study ID ST002097, and three additional rows shared analysis IDs. After this step, the snapshot contained 4,499 studies and 7,367 analyses. The rule makes identifiers unique; it does not show that rows sharing an identifier had identical content.

The species endpoint returned 4,773 rows representing 4,770 unique study-Latin-name pairs across 4,483 study IDs. Among those studies, 4,395 listed one species and 88 listed more than one. Eighteen studies in the summary index had no species-index row, and two species-index IDs had no matching summary row. Species percentages use the 4,483 annotated studies as the denominator. They can sum above 100% because a study may include more than one species.

Rank	MW source label	Studies	Share of annotated studies
1	Homo sapiens	1,852	41.31%
2	Mus musculus	1,463	32.63%
3	Rattus norvegicus	257	5.73%
4	Escherichia coli	54	1.20%
5	Plasmodium falciparum	53	1.18%
6	Drosophila melanogaster	32	0.71%
7	Caenorhabditis elegans	27	0.60%
8	Arabidopsis thaliana	25	0.56%
9	Danio rerio	24	0.54%
10	Saccharomyces cerevisiae	24	0.54%

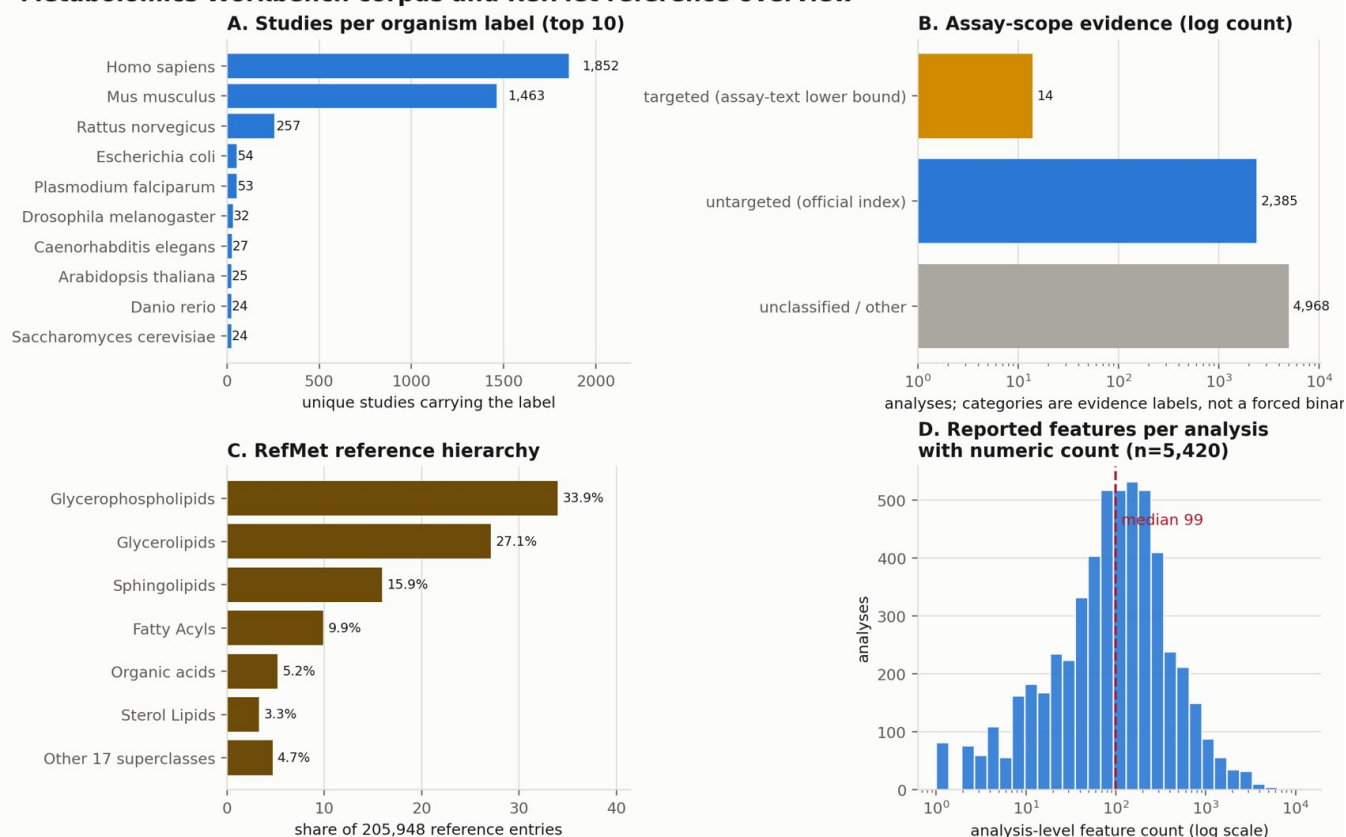
Targeted, untargeted, and unresolved assay scope

MW provides a dedicated untarg_studies index but no matching targeted-only bulk index. The untargeted index identifies 2,385 analyses in 1,250 studies. A conservative search of the analysis text finds 14 explicitly targeted analyses in 14 studies. The remaining

4,968 analyses are unclassified or other; the available metadata do not support relabeling them as targeted. At the study level, 3,235 studies are neither explicitly targeted nor present in the untargeted index, or they have no analysis row. The current evidence therefore does not support a complete binary targeted/untargeted classification.

The number_of_metabolites endpoint returned 5,485 records: 5,420 numeric values and 65 blanks. Another 1,882 analyses had no count record. Numeric values ranged from 1 to 11,992, with a median of 99, and summed to 1,231,376 analysis-feature instances. This is not a unique-metabolite count. The same chemical can recur across analyses, and an untargeted feature may not be an identified compound.

Metabolomics Workbench corpus and RefMet reference overview



Counts are frozen from the public MW bulk REST indexes on 2026-08-24 UTC. RefMet entries are nomenclature records, not study-observed or MSI level-1 identities.

Figure 1. MW corpus overview. Species counts are study incidences, assay categories are analyses, and feature counts are analysis-level reported values. RefMet percentages describe a separate reference nomenclature snapshot.

2. RefMet chemical hierarchy

Here, superclass/class/subclass refers to the local RefMet hierarchy (super_class, main_class, sub_class), not a PubMed or PubChem classification. The snapshot contains 205,948 unique nomenclature entries, including 205,919 with a nonblank formula, across 23 superclasses, 333 main classes, and 960 subclasses. The local source file does not record a release identifier. These rows are reference labels. They are not 205,948 chemicals shown to occur in MW, and a shared formula does not establish identity.

Rank	Superclass (n)	Main class (n)	Subclass (n)
1	Glycerophospholipids (69,886)	Triradylglycerols (33,688)	TAG (33,431)
2	Glycerolipids (55,850)	Glycosphingolipids (16,592)	FAHFA (14,719)
3	Sphingolipids (32,811)	Fatty esters (15,818)	HexCer (10,579)
4	Fatty Acyls (20,340)	Glycerophosphoglycerols (15,387)	Tripeptides (8,017)
5	Organic acids (10,648)	Glycerophosphoethanolamines (12,424)	O-DG (5,770)

The three largest superclasses account for 76.98% of the entries, so the reference file is heavily weighted toward lipids. That pattern describes RefMet itself. Applying it to MW observations would first require a study-analysis-feature crosswalk and supporting identity evidence.

3. Worked comparisons: MoTrPAC and Lac-Phe

Acute human MW-MoTrPAC comparison

We compared MW study ST004303 (266 effect rows; 261 case/space-normalized labels) with the cached public MoTrPAC human acute-endurance late-post-versus-pre plasma effect table (1,657 rows; 1,509 normalized labels across 11 targeted and untargeted panels). The screen found 99 exact case/space-normalized label overlaps and 102 punctuation-insensitive candidate name-key overlaps. Duplicate labels or panels made 46 of the 102 candidates ambiguous. The other 56 were unique in both tables at the duplicate-key screening gate. Neither table carries source-stable RefMet identifiers, so none of these candidates is treated as structurally validated or ready for pooling. They form a review queue; no meta-analysis was run.

Lac-Phe worked example

An exact RefMet-name query for N-Lactoyl phenylalanine returned 85 raw MW metabolite-statistics rows; the variants Lac-Phe and N-lactoylphenylalanine returned none. After repeated study-analysis pairs were removed, 60 label-associated study-analysis pairs from 57 studies remained, including 24 analyses from 22 human studies. The resolver's database annotation for N-Lactoyl phenylalanine is RefMet RM0131640 (formula C12H15NO4; PubChem 11075454). This annotation does not prove the identity of the focal feature. The public source we inspected did not document linked authentic-standard retention time plus MS/MS evidence for that feature. It therefore stays in the human-review queue and is not assigned MSI level 1.

For the study-reported whole-blood feature labeled Lactoyl Phenylalanine in ST003662, 116 of 137 participant rows had complete positive-abundance pre/post values. Ninety-two of the 116 increased. The acute post-minus-pre analysis used participant-level $\log_2(\text{post}) - \log_2(\text{pre})$ differences and estimated a mean paired $\log_2$ fold change of +1.1586 (geometric mean paired fold 2.23x), with $p = 2.73\text{e-}15$ and BH FDR = $2.50\text{e-}14$. This was the fourth-largest $\log_2$ fold change among 165 usable all-athlete estimates. In the study-reported whole-blood units, arithmetic means rose from 0.539 to 0.755 micromol/L, a ratio of 1.40. That ratio and the paired log-scale estimate answer different questions and should not be interchanged. The inspected source did not encode protocol modality, intensity, duration, or the post-exercise sampling delay.

The RefMet snapshot contains 22 labels beginning with N-Lactoyl. MW metabolite statistics associate 17 of them with at least one study-analysis pair. Only the Lac-Phe label is associated with ST003662, and none of the 22 labels appears in the inspected

cached public human acute-plasma or rat MoTrPAC tables. We record the missing labels as a coverage gap. Their biological effects remain unknown.

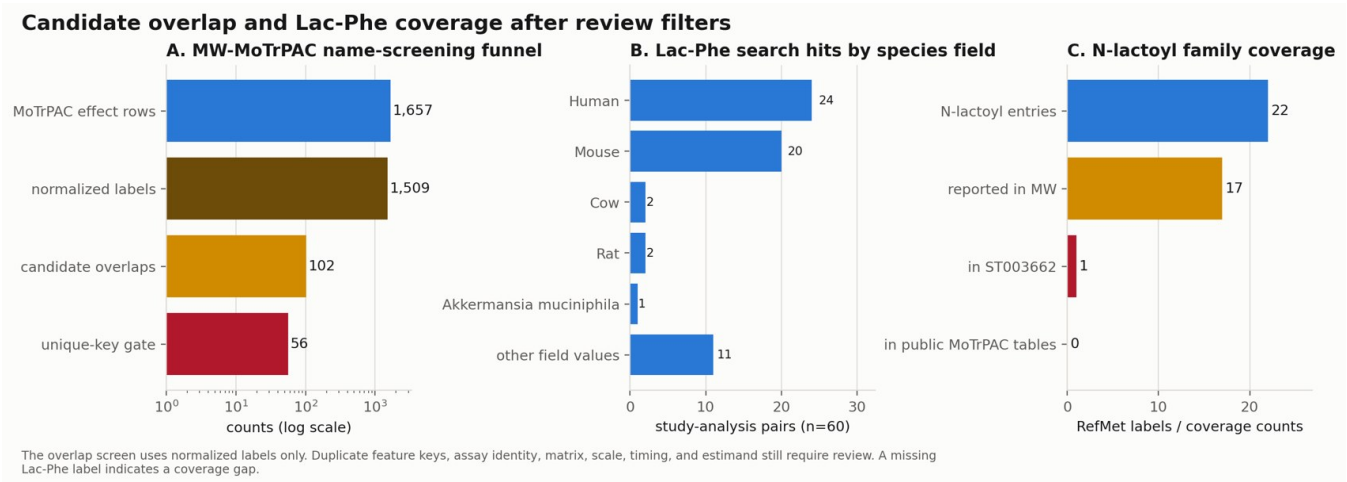


Figure 2. Review filters applied to the MW-MoTrPAC name screen and the Lac-Phe example. Fifty-six name-key candidates remain unambiguous, but their chemical identity is not verified. The Lac-Phe RefMet label is associated with several MW-reported species fields, and N-lactoyl coverage differs across RefMet, MW, the focal study, and cached MoTrPAC tables.

4. Current capabilities

Deterministic Python code runs the core workflows. Agent and skill contracts define what each role may do and when human review is required. The current inventory has 22 canonical roles and 24 logical project skills; mirrored Codex and Claude copies count once. The roles assign responsibility for source retrieval, metadata and identity review, statistical review, plotting, and pipeline planning.

Workflow	Reviewable output	Scientific safeguard
Discovery	Inclusion criteria, ranked studies, literature and repos...	Missing repository support is flagged for review
Metadata extraction	Study, dataset, variable, timing, assay, and provenance ...	Unknown stays unknown; it is not converted to absent
Identity and harmonization	Identity candidates, confidence-scored crosswalks, assay...	Similar names do not authorize a merge or pooling
Comparison and plotting	MoTrPAC-style effects, overlap screens, descriptive RefM...	Descriptive overlap is not reported as replication or pa...
Review and reporting	Queues, advisory domain packets, manifests, scored readi...	Domain packets stay advisory; mappings under review prod...

The source registry contains 28 sources across 27 retrieval lanes. Some lanes have live core adapters; others use offline snapshots, optional life-science connectors, standards references, or planned/manual routes. Registration records where a request should go. It does not prove that live retrieval works. The current synthetic/local release gate ran 150 tests and all passed. In the synthetic benchmark, candidate precision/recall and review capture were 1.00, unsafe auto-accept was 0/12, and quality-score agreement within 0.15 was 8/10. These results check reproducibility on constructed fixtures, not performance on an independent external dataset.

The current implementation does not reconcile chemical structures automatically across every source. It leaves unreported features as unknown and blocks pooling until matrices, assays, units, scales, time points, protocols, and estimands are compatible. Several registered sources still lack executable adapters. The supporting files are the frozen MW manifest, derived evidence bundle, skill audit, scientific-readiness audit, and full Lac-Phe dossier.

Summary

The MetaboTyping RA Agent Workbench helps research teams find, curate, and compare metabolomics datasets without hiding the judgment calls. The current worktree has 22 canonical agent roles and 24 project skills mirrored across Codex and Claude; two older role names remain only as aliases. The workflow turns a research question into inclusion criteria, ranks candidate studies, keeps repository records and provenance intact, and produces study, dataset, and variable cards. It can also flag uncertain metabolite identities, propose confidence-scored crosswalks, draft assay-aware harmonization plans, assess dataset readiness, review literature, and render MoTrPAC-style figures and review packets. Shared names do not justify a merge. Uncertain mappings go to human review, and pooling requires both compatibility evidence and human approval.

The frozen MW snapshot contains 4,499 study summaries and 7,367 analyses. Species metadata cover 4,483 study IDs and 4,770 study-organism incidences across 481 source labels. MW identifies 2,385 analyses in 1,250 studies as untargeted. Only 14 analyses are explicitly labeled targeted in assay text; the other 4,968 remain unclassified and cannot be relabeled targeted from the available evidence. The local RefMet file is separate. It contains 205,948 nomenclature entries across 23 superclasses, 333 main classes, and 960 subclasses, not 205,948 metabolites observed in MW. The acute human screen found 102 punctuation-insensitive candidate name-key overlaps between MW ST004303 and the cached public human acute-endurance late-post-versus-pre plasma effect table; 56 were unique on both sides at the duplicate-key screening gate. The exact RefMet-name search returned Lac-Phe label associations for 57 studies and 60 study-analysis pairs. In ST003662, the whole-blood feature labeled Lactoyl Phenylalanine increased by a mean paired log2 fold change of 1.16 across 116 complete pairs. The inspected cached public human acute-plasma and rat MoTrPAC tables did not contain that label. The missing label is treated as unmeasured coverage, not a null effect. Identity and pooling remain under review.

Provenance note. MW counts came from documented MW REST v1.2 bulk endpoints through the repository's allowlisted fetcher. The frozen files include raw payloads, normalized tables, SHA-256 hashes, endpoint URLs, timestamps, row counts, and de-duplication rules. RefMet, Lac-Phe, MoTrPAC, agent/skill, and test counts come from local snapshots. Their source files and derived tables are listed in data/report_metrics.json and the adjacent CSV bundle.